Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages & Databases: Python, SQL, Java, C/C++, R, SQLite, MySQL, MongoDB
Data Analytics & Engineering: Data Cleaning, ETL, Exploratory Data Analysis, Feature Engineering, Data Transformation, Data Modeling, Relational Database Design, Data Validation, Pandas, NumPy
Machine Learning & AI: Scikit-learn, PyTorch, TensorFlow, Random Forest, XGBoost, Isolation Forest, LSTM, Time-Series Analysis, Anomaly Detection, Generative AI, RAG
Data Visualization: Power BI, DAX, Tableau, Matplotlib, Plotly, Streamlit
Software Engineering & Tools: FastAPI, Flask, REST APIs, Git/GitHub, PyTest, Jupyter Notebook

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** — ***Aug – Sep, 2025

- Analysed 2 real-world business datasets containing 8,500+ records, applying data preprocessing and AI-assisted analytics to identify trends and generate 15 actionable business insights for data-driven decision-making.
- Validated analytical findings for accuracy and consistency, reviewing outputs to ensure reliable and meaningful insights for business recommendations.
- Established a standardised analytics workflow covering data preprocessing, insight generation, and validation, enabling consistent conversion of raw business data into actionable recommendations.

PROJECTS

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered an end-to-end Python analytics pipeline processing 50K+ PaySim transactions, incorporating data validation, duplicate detection, statistical outlier analysis, vendor-velocity checks, HHI concentration analysis, and Isolation Forest.
- Designed a star-schema SQLite data warehouse and implemented a composite 0–100 risk-scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Developed a 5-page Power BI dashboard with 20+ DAX measures to deliver vendor risk, anomaly-rate, and transaction trend insights, supported by automated pipeline tests.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built a full-stack receipt processing platform that extracted structured data from 12,000+ receipts with approximately 96% OCR accuracy using EasyOCR and OpenCV.
- Developed secure FastAPI backend services with JWT authentication, Google OAuth, and REST APIs, reducing average API response time to under 180 ms.
- Automated expense categorisation and analytics through NLP-based processing, reducing manual bookkeeping effort by 80% while providing interactive financial dashboards.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Engineered a real-time market analytics pipeline processing 2M+ NSE records, combining price signals and financial-news sentiment to detect unusual market behaviour.
- Developed and evaluated Isolation Forest and LSTM Autoencoder models, achieving 88% precision and 85% recall on validation data for anomaly detection.
- Optimised the analytics pipeline to reduce detection latency by 35%, enabling near real-time monitoring and faster identification of unusual market events.

***Customer Lifetime Value Engine* |LINK**

- Built a probabilistic model (BG/NBD + Gamma-Gamma) on 1M+ historical transaction records to forecast customer lifetime value directly analogous to modeling player value and monetization potential.
- Segmented customers into value-based cohorts to identify which segments drive the most revenue, supporting prioritization decisions.
- Identified the top 20% of customers driving ~65% of forecasted revenue, visualized through Power BI dashboards for stakeholder-facing reporting.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Deloitte Australia Data Analytics Job Simulation |**LINK** 2025
- Accenture Software Engineering Job Simulation | **LINK** 2025